

AlBaraa AlOlabi

Computer Vision Engineer

Abu Dhabi, UAE • +971 54 328 0983 • 666645@gmail.com • [LinkedIn](#) • [GitHub](#) • [HuggingFace](#) • albaraaalolabi.dev

SUMMARY

Computer Vision & Robotics Engineer delivering production-grade Edge AI and Autonomous Systems. Expert in optimizing deep learning models for on-device inference (ONNX, PyTorch Mobile) and building agentic architectures. Proven track record in safety-critical perception systems, multi-agent frameworks, and real-time optimization for autonomous applications.

EDUCATION

AI Ain University

Bachelor of Science in Computer Science – College of Engineering

Abu Dhabi, UAE

Sep 2022 – Expected Jun 2026

RESEARCH & PUBLICATIONS

Author: "The Impact of Artificial Intelligence in Education on Student Learning Outcomes and Teaching Methods" [Poster](#) — [Link](#)

IEEE SNAMS 2025 – Vienna, Austria

- Sole-authored research paper accepted to **IEEE SNAMS 2025** (Scopus & DBLP indexed).
- Presented on 27 November 2025 and publication in IEEE Xplore.

EXPERIENCE

Cellula Technologies

Computer Vision Engineer Intern

UAE

Jan 2026 – Present

- Implemented a custom ResNet-inspired CNN from scratch in PyTorch for dental disease classification, achieving 97.67% accuracy across 7 categories (e.g., Caries, Oral Cancer) using advanced data augmentation techniques.
- Collaborating with cross-functional teams on end-to-end CV pipelines including data preprocessing, model training, evaluation, and deployment.

Samsung Innovation Campus – AI & ML Program

Samsung Selected Participant – Advanced training in ML, DL, NLP, and AI systems

UAE

Sep 2025 – Dec 2025

- Engineered a safety-critical hazard detection module for assistive navigation, achieving 85% accuracy in identifying environmental obstacles (Safe/Caution/Unsafe) using transfer learning on MobileNetV2.
- Designed a robust data pipeline processing 3,000+ environmental samples with class-balancing and augmentation, simulating real-world unstructured environments for reliable perception.

PROJECTS

Mission Control MCP (Autonomous System Architecture) – [Demo](#)

Enterprise-Grade Agentic Framework

Python, FAISS, RAG, System Design

- Architected a modular Model Context Protocol (MCP) server providing "nervous system" capabilities for AI agents, including memory (RAG/Vector Search), perception (Web/Doc parsing), and analysis.
- Implemented robust vector retrieval using FAISS and Sentence Transformers, enabling long-term state persistence and context-aware execution for autonomous workflows.

Global TomatoCare (Edge AI & On-Device Inference) – [Github](#)

Offline-First Plant Disease Detection System

PyTorch Mobile, ONNX, Quantization

Graduation Project

- Developed a lightweight diagnostic agent optimized for edge deployment, capable of running fully offline on mobile hardware using quantized MobileNetV2.
- Engineered an efficient inference engine utilizing ONNX Runtime, achieving 150ms latency for real-time plant disease classification across 9 pathogen categories.

CleanCity Agent (Autonomous Planning System) – [Demo](#)

Autonomous Resource Allocation Agent

YOLOv8, Planning Algorithms

- Built an autonomous planning agent that perceives unstructured environments via YOLOv8 and generates executable resource allocation plans for waste management.
- Integrated vision-based state estimation with Large Language Models (Gemini) to convert visual data into structured action plans (Cleaning Reports).

SKILLS

Languages: Python, C++, C, Java, JavaScript

AI/ML Frameworks: PyTorch, TensorFlow, Scikit-learn, HuggingFace, NumPy, Pandas

Robotics & Perception: ROS/ROS2 (Basics), SLAM, Object Detection (YOLO), Semantic Segmentation, Sensor Fusion

Edge AI & Optimization: ONNX, TensorRT, PyTorch Mobile, Model Quantization, Real-time Systems (30+ FPS)

Core Stack: Python, C++, OpenCV, Docker, Linux, Git, MCP, Vector Databases (FAISS)

Core Competencies: Computer Vision Pipelines, System Integration, Model Deployment, Rapid Prototyping, Technical Research

CERTIFICATIONS

CS50x: Introduction to Computer Science – [link](#)

CS50P: Introduction to Programming with Python

CS50AI: Introduction to Artificial Intelligence with Python

AI Foundations: Machine Learning

Harvard University — Sep 2024

Harvard University — May 2025

Harvard University — Present

LinkedIn Learning — May 2025